

How to generate the new format of γ_k grid files

- 1) Use $w_asym.f$ and $w_pert.f$ files to generate L_0 and A_3 to α_s^1 and α_s^2 order.
- 2) Use legacy or ResPos2 code to generate L_0 and A_{0-4} to α_s^1 order, which is the $-\gamma_-$ file.
- 3) The new format of $-\gamma_k$ file, up to $\alpha_s^{(2)}$, is to directly use the α_s^2 output from $w_pert.f$

$$L_0 \left\{ 1, \frac{A_3}{L_0}, \frac{A_{0-4}}{L_0} \right\} \quad \begin{matrix} \alpha_s^2 & \alpha_s^2 & \alpha_s^2 & \alpha_s^2 \end{matrix} \quad R_{A_{0,1,2,4}} \equiv \frac{\frac{A_i^{(2)}}{L_0^{(2)}}}{\frac{A_i^{(1)}}{L_0^{(1)}}}$$

where $\left(\frac{A_i}{L_0} \right)^{(2)} = \underbrace{R_i}_{\text{from R-grid file}} * \underbrace{\left(\frac{A_i}{L_0} \right)^{(1)}}_{\text{from } -\gamma_- \text{ grid}}$

4) Likewise, the $-\gamma_-$ grid file gives

$$L_0 \left\{ 1, \frac{A_3}{L_0}, \frac{A_{0-4}}{L_0} \right\} \quad \text{at } \alpha_s^{(1)}.$$

$\alpha_s^{(1)} \quad \alpha_s^{(1)} \quad \alpha_s^{(1)}$

- 5) We need to add an additional comment line to
 - ① indicate the pbf's used, and
 - ② distinguish different version of the grid files.